Statistical analysis of experimental data
Concept of probability

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FACULTY OF PHYSICS
UNIVERSITY OF WARSAW

Lecture 02

October 15, 2026

Concept of probability

1. Introduction
2. Basic terms
3. Definition of Probability
4. Properties of Probability
5. Bayes' Theorem
6. Homework

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Statistical analysis is also required for proper experiment design.
It is similarly important for interpretation of "single measurements"...

Statistical analysis of experimental data

The goal of this lecture is to present basic concepts and methods of statistical analysis of experimental data...

We will quite often refer to problems encountered in the elementary particle physics, as this is the perfect test ground for statistical analysis

This is because all particles (of the same type) are identical.
We can easily assure same initial conditions for the repeated measurement...
This is often a fundamental problem in social or medical sciences...

The knowledge of particle physics is not required (but for some fundamental concepts like the invariant mass or mean lifetime) and the presented analysis methods can clearly be used also in other fields of science...

Typical problems adapted from book by S.Brandt

- We measure properties of the selected individuals from a large population (could be elementary particles of a given type).
  What is the precision of the measurement?
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  When can we claim that given theoretical model is excluded?

- A general model describing the process studied is known, but parameters of this model must be obtained from experiment (very common case in particle physics).
  What is the optimum procedure for extracting model parameters from the data? How can the experiment be optimised to give strongest constraints on the model parameters?

About myself

I finished Physics studies at this faculty in 1987. For my MSc I analysed data stored on magnetic tapes writing the code in FORTRAN on punched cards...

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- analysis of electron-proton scattering events in ZEUS experiment at HERA
- analysis of astronomical images in a search for short optical bursts in the sky (connected to Gamma Ray Bursts, gravitational wave sources etc.) - Pi of the Sky experiment
- prospects for new particle searches in future electron-positron colliders (ILC and CLIC)
- optimization of sampling calorimeter design for future experiments

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One of my goals at this lecture is to share some of my experience with you...

Course plan

14 lectures, Thursdays, 9:15 – 12:00, room B2.38 (with two breaks)
From October 8th, 2026 to January 21th, 2027

Homework exercises
Solutions have to be uploaded to Kampus within two weeks!
Until midnight before the lecture! Proposed solutions will be presented at the lecture.

Written exam, five problems to be solved
Solutions to be uploaded to Kampus within one week!

Assessment criteria

Minimum of 50% of points collected from exercises and exam (with the same weight).

Make-up assessment: 50% of points from written examination only.

Problem solutions Apply for all homework and exam problems.

Solutions should be uploaded to Kampus as PDF documents (scan/photo of hand-written solution is also OK) including description/justification of the approach used, numerical results and relevant plots, as well as result discussion.

The Python notebook can be uploaded as a second file
or the link to notebook in Google Colab can be added as a comment.
Make sure your code runs "from scratch" (no hidden dependencies) and Google Colab accessible.

By uploading the solutions to Kampus (homework and exam) you declare that they resulted from your own work and that you have not obtained, shared nor discussed them with anyone. This includes AI...

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Use of AI is allowed on purely technical level, to solve problems faced when writing the code, but not for the actual problem solution...

Web resources

Kampus platform will be used for lecture slides, home exercises and final exam:

https://kampus-kursy.ckc.uw.edu.pl/course/view.php?id=10840

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All scripts discussed at the lecture will be available for download from the github repository:

https://github.com/zarnecki/SAED

Inserted in the lecture slides are links to open them in your browser using the Google Colab.

You can also use this link 

to open repository and select the notebook to load.

Please remember that you need to save the file to (your) Google Disc before you make any modifications. Otherwise, all your changes will be lost!..

Some references

- G. Bohm and G. Zech, Introduction to Statistics and Data Analysis for Physicsts, Verlag Deutsches Elektronen-Synchrotron, 3rd edition;
- S. Brandt, Data Analysis: Statistical and Computational Methods for Scientists and Engineers, Springer 2014;
- **M. Bonamente, Statistics and Analysis of Scientific Data**, Springer 2017;

- R.J. Barlow, Practical Statistics for Particle Physics, PDF from arXiv;
- Max Bramer, Principles of Data Mining, Springer 2016;
- David Forsyth, Probability and Statistics for Computer Science, Springer 2018;
- Particle Physics Reference Library Volume 2 (chapter 15), Springer 2020.

Course organization

If you have any special needs, requests or problems related to the course, please let me know as soon as possible.

You can stay after the lecture, come to my office (4.01) or just send an email (to zarnecki@fuw.edu.pl)

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For other problems

- University website dedicated to equality: https://rownowazni.uw.edu.pl/en/
- Academic Ombudsperson: https://ombudsman.uw.edu.pl/en/start/
- Representative for Equality at our Faculty: Magdalena Popielska
- Psychological Support Centre (CPP): https://cpp.uw.edu.pl/en/home-page/
- Office for Persons with Disabilities (BON): https://bon.uw.edu.pl/en/home/

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To allow for statistical analysis we make many measurements...

However, there are always some random factors and results of subsequent measurements are usually different. It is important to understand where these variations come from!

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   These fluctuations are unavoidable, we can not reduce them.
   But usually this is also the most interesting case for us...

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   We can try to reduce the fluctuations (and thus improve precision)
   by adjusting the measurement procedure...

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   Examples: measurement of the drug effectiveness, answers to questionnaires in various types of surveys (eg. asking for age), mass spectrometry, composition and energy spectra of cosmic rays...

   Results will usually depend on the way the tested population sample is selected.
   This selection has to be well defined!

Experiments

It is crucial to correctly identify the source of fluctuations!

Example: coin diameter measurement.

Fluctuations in numerical results can be due to:

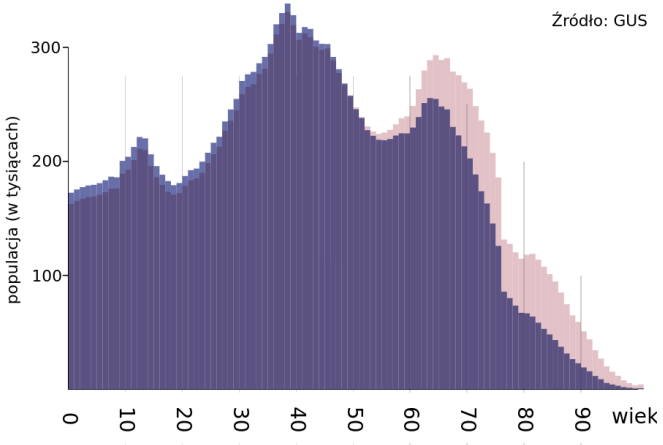
- finite precision of the instrument there is always some measurement error
- the measurement method, how we define the diameter
  in particular when the coin is not exactly round
- fluctuations in the actual coin size,
  if we measure a set of coins, not a single one

$\Rightarrow$ we need to define the problem properly!

In addition to measurement fluctuations, we also need to consider possible systematic effects.
We will come back to this later...

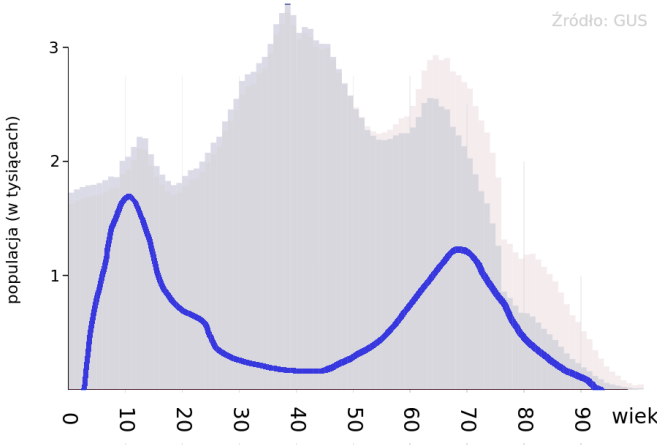
Experiments Example: influence of the test sample choice.

Age structure of the population of Poland (GUS estimate for 2022, men and women).



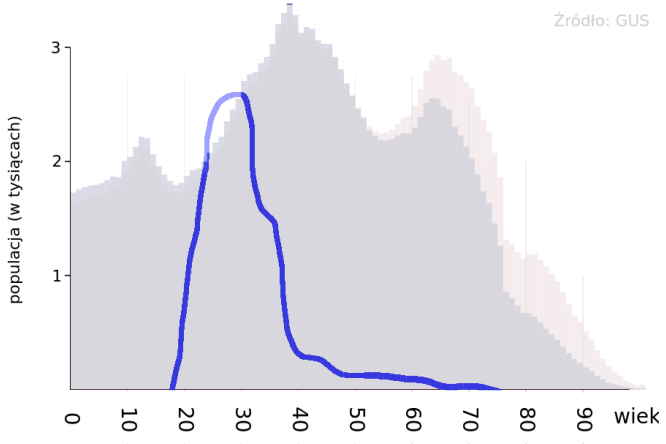
Experiments Example: influence of the test sample choice.

Possible result for the survey in a public park (on working day).



Experiments Example: influence of the test sample choice.

Possible result for the survey at the university campus.



Elementary event

Outcome of a single experiment (measurement):

- result of the roll of a die
- observation of N charged particles in the particle collision
- observation of nuclear decay after given time
- observation of given process, eg. decay $K^+ \to \mu^+ + \nu_\mu$

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Roll of a die

Six possible outcomes in the sample space: $\Omega = \{1, 2, 3, 4, 5, 6\}$

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Charged particle multiplicity

Particle counting gives non-negative integer number: $\Omega = \{0, 1, 2, \ldots\}$

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Decay time measurement

Time interval measured is a real number: $\Omega = \mathbb{R}_+$

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Observation of K^+ decay:

Sample space should include all possible (observable) decay channels:

$$\Omega = \{K^+ \to \pi^+\pi^\circ, K^+ \to \pi^+\pi^\circ\pi^\circ, K^+ \to \pi^+\pi^+\pi^-, \quad K^+ \to e^+\nu_e, K^+ \to \mu^+\nu_\mu,$$

$$K^+ \to \pi^\circ e^+\nu_e, K^+ \to \pi^\circ\mu^+\nu_\mu, K^+ \to \pi^\circ\pi^\circ e^+\nu_e, K^+ \to \pi^+\pi^- e^+\nu_e, \quad K^+ \to \pi^+\mu^+ e^-\}$$

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Examples of events

Roll of a die

- six: $A_1 = \{6\}$
- odd number: $A_2 = \{1, 3, 5\}$
- even number: $A_3 = \{2, 4, 6\}$
- any number: $A_4 = \Omega = \{1, 2, 3, 4, 5, 6\}$

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Charged particle multiplicity

- pair production: $A_1 = \{2\}$
- odd number: $A_2 = \{1, 3, 5, \ldots\}$
- even number: $A_3 = \{2, 4, 6, \ldots\}$

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Observation of K^+ decay:

- Hadronic decays

$$A_1 = \{K^+ \to \pi^+\pi^\circ, K^+ \to \pi^+\pi^\circ\pi^\circ, K^+ \to \pi^+\pi^+\pi^-\}$$

- Leptonic decays

$$A_2 = \{K^+ \to e^+\nu_e, K^+ \to \mu^+\nu_\mu\}$$

- LFV decay (forbidden in SM)

$$A_3 = \{K^+ \to \pi^+\mu^+e^-\}$$

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- mutually exclusive events: two events with no common outcome
  $\Rightarrow$ their intersection is an empty event: $A \cap B = \emptyset$

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Classical definition as developed in the 18th–19th centuries

If the sample space contains N_Ω elementary events (possible outcomes of the experiment) and the considered event A contains N_A elementary events, then, **assuming all elementary events are equally probable**

$$P(A) = \frac{N_A}{N_\Omega}$$

Classical definition - example

see also in Wikipedia

In a straight poker game, consider player dealing five cards from a deck of 52.
Sampling space contains all possible "hands", all possible sets of 5 cards selected out of 52 (order is not relevant):

$$N_\Omega = \binom{52}{5} = 2\,598\,960$$

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We have four suits and the sequence can start from 9 numbers (from 1 to 9):

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What is the probability to get four cards of the same rank?

$$N_{four} = 13 \times 48 = 624 \Rightarrow p_{four} = \frac{624}{2\,598\,960} \approx 2.4 \cdot 10^{-4}$$

where we had to take into account the number of options (48) for the fifth card !

Classical definition

The classical definition of the probability works fine in many simple problems, the gambling games in particular.

However, the sampling space and elementary events have to be uniquely defined!

This is usually not a problem for experimental results given by discrete numbers (eg. roll of a die, card games, etc.).

But problems arise when this approach is to be applied to experiments with continuous spectra of possible measurement results...

This is well illustrated by the "Bertrand paradox"

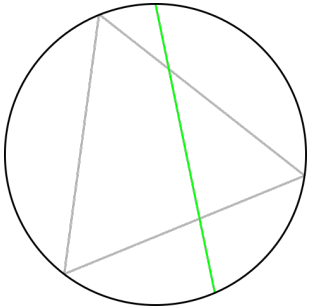
Bertrand paradox

description adapted from R. J. Barlow
pictures from Wikipedia

Definition of the problem:

In a circle of radius R an equilateral triangle is drawn.

What is the probability that the length of a random chord is greater than the triangle side?



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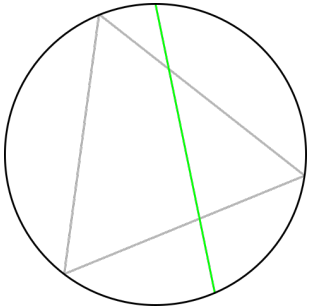
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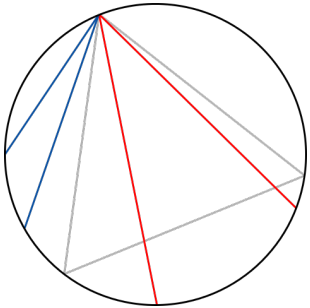
What is the probability that the length of a random chord is greater than the triangle side?

Solution 1:

A random chord can be defined as a line connecting two random points on the circle.

Without loss of generality, we can move one of its ends to the vertex of the triangle.

The chord will be longer than the side of the triangle, if its other end is between the two other vertices $\Rightarrow$ probability is $1/3$



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What is the probability that the length of a random chord is greater than the triangle side?

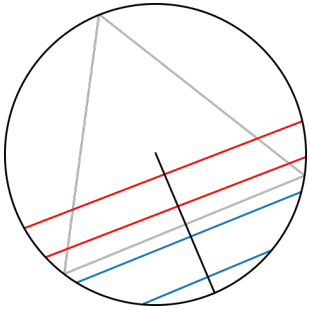
Solution 2:

Without loss of generality, we can rotate a random chord in such a way that its centre is on the indicated radius of the circle.

The chord will be longer than the side of the triangle, if its centre is inside the triangle.

The side of the triangle cuts the radius in the middle

$\Rightarrow$ probability is $1/2$



Bertrand paradox

description adapted from R. J. Barlow
pictures from Wikipedia

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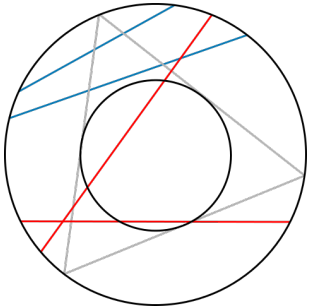
Solution 3:

We can draw a chord by choosing a random point in the circle as its centre (unique definition of the chord).

The chord will be longer than the side of the triangle, if its centre is inside the circle of radius $R/2$.

The surface of the circle is proportional to radius squared

$\Rightarrow$ probability is $1/4$.

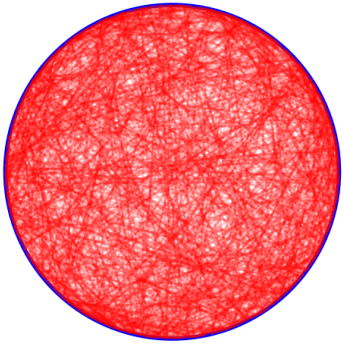


Bertrand paradox

02_Bertrand.ipynb Open in Colab

It is useful to visualise the problem by plotting a number of random chords

Solution 1:



Probability from 100'000 trials:

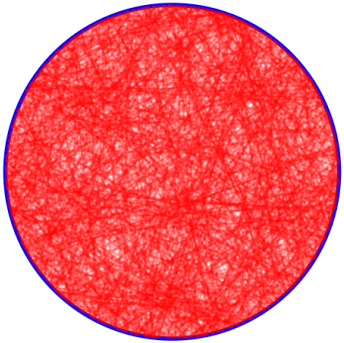
0.3330 +/- 0.0015

Bertrand paradox

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Solution 2:



Probability from 100'000 trials:

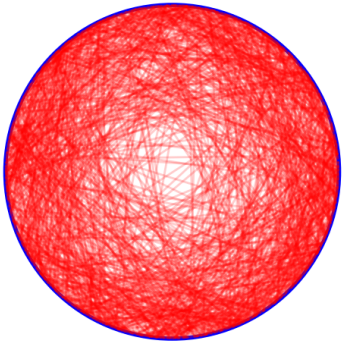
0.4994 +/- 0.0016

Bertrand paradox

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It is useful to visualise the problem by plotting a number of random chords

Solution 3:



Probability from 100'000 trials:

0.2509 +/- 0.0016

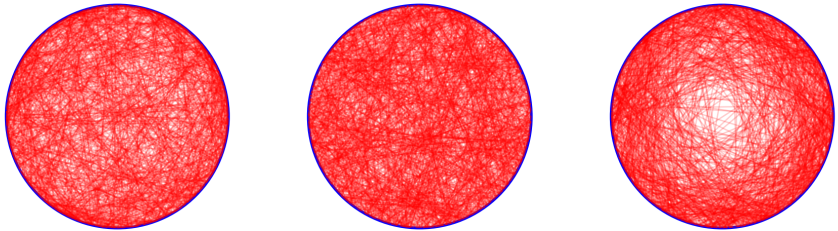
Bertrand paradox

Base on the classical probability definition, we can get three contradictory answers! Why?
For discrete elementary events we could just assume they are equally probable...

Bertrand paradox

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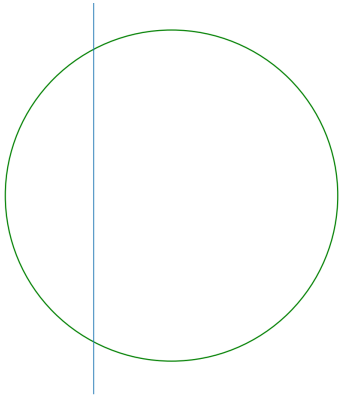
In the considered continuous sample space, **"equally probable" elementary events** are not uniquely defined!. We need to define how the actual experiment is performed...



Different answers correspond to different assumptions on the chord distribution...

Bertrand paradox - my preferred solution

The problem can be reduced to the definition of the random chord distribution...
Each cord corresponds to a straight line on the plane - can we define a random line?

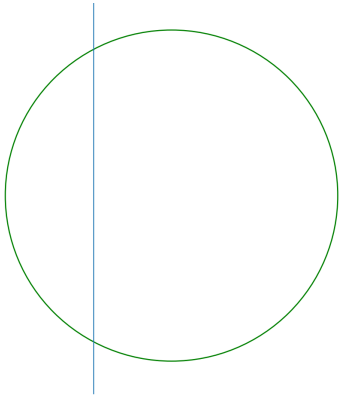


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We can revert a problem: fix a line and consider random circle position

Only perpendicular direction matters, position can change from $-R$ to $+R$ (line has to cross the circle)



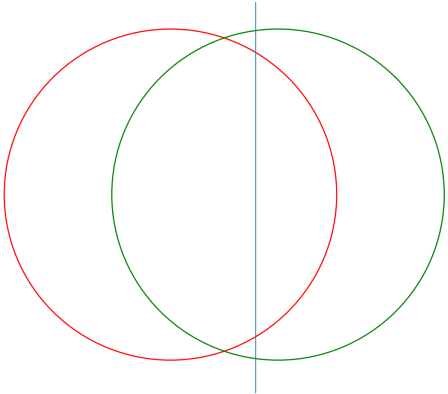
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Chord is longer than $\sqrt{3}\mathrm{R}$, if position within $\pm\frac{1}{2}\mathrm{R}$ (see solution 2)



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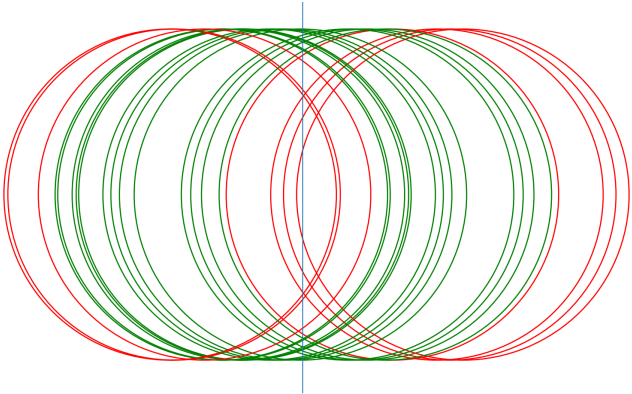
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Chord is longer than $\sqrt{3}\mathrm{R}$, if position within $\pm\frac{1}{2}\mathrm{R}$ (see solution 2)

Uniform circle position on the surface corresponds to solution 2 ($p = \frac{1}{2}$)



Definition of Probability

Frequentist definition of probability

When repeating the same experiment a large number of times, $N \gg 1$, the probability of A

$$P(A) = \lim_{N \to \infty} \frac{N(A)}{N}$$

where $N(A)$ is the number of occurrences of the event A

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Sample space Ω reflects the way the experiment is done. Experimental procedure has to include definition of elementary events constituting the sample space.

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While this is not directly visible in the formula, the probability still depends on the considered sample space Ω.

Sample space Ω reflects the way the experiment is done. Experimental procedure has to include definition of elementary events constituting the sample space.

"equally probable" elementary events do not need to be defined, but only the procedure for repeating the experiment (!).

Definition of Probability

The frequentist (sometimes also called "classical") definition of probability gives direct recipe for the analysis of experimental data...

However, it is not always possible to perform the experiment a very large (infinite) number of times... We need some additional guidance to know how to define the probability.

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Kolmogorov Axioms

Kolmogorov (1933) formulated the three conditions
which have to be fulfilled by probability $P(A)$ of an event $A \subset \Omega$:

1. probability is a non-negative number: $P(A) \geq 0$
2. probability of all possible outcomes (sample space): $P(\Omega) = 1$
3. if A and B are mutually exclusive events: $P(A \cup B) = P(A) + P(B)$

We can derive all properties of the probability from these three axioms...

Concept of probability

1 Introduction

2 Basic terms

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Fundamental properties

Following properties can be derived from the Kolmogorov axioms:

- probability of the empty event is zero:

$$P(\emptyset) = 0$$

$\emptyset$ and Ω are mutually exclusive and $\Omega = \Omega \cup \emptyset$

$\Rightarrow P(\Omega) = P(\Omega) + P(\emptyset) \quad \Rightarrow P(\emptyset) = 0$

Fundamental properties

Following properties can be derived from the Kolmogorov axioms:

- probability of the empty event is zero:

$$P(\emptyset) = 0$$

- probability of complementary event:

$$P(\bar{A}) = 1 - P(A)$$

A and $\bar{A}$ are mutually exclusive and by definition $A \cup \bar{A} = \Omega$
$\Rightarrow P(A) + P(\bar{A}) = P(\Omega) = 1 \quad \Rightarrow P(\bar{A}) = 1 - P(A)$

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- probability of the union of two events:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Union of A and B can be decomposed as:

$$P(A \cup B) = P(A \cap \bar{B}) + P(B \cap \bar{A}) + P(A \cap B)$$

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$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

and then one can note that

$$P(A \cap \bar{B}) = P(A) - P(A \cap B) \quad \text{and} \quad P(B \cap \bar{A}) = P(B) - P(B \cap A)$$

Statistical Independence

Two events A and B are said to be statistically independent if and only if

$$P(A \cap B) = P(A) \cdot P(B)$$

Two important properties follow:

- mutually exclusive (nonempty) events cannot be independent
- if A is subset of B, $A \subset B$, they cannot be independent, unless $B = \Omega$

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Conditional Probability

When two events are not independent, we can consider probability of event A given that another event B is observed:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{or 0 if } P(B) = 0$$

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For independent events: $P(A|B) = P(A)$

Example (1)

Rolling a single die: $N = 6$ possible outcomes (elementary events).
We assume each outcome has the same probability $\Rightarrow p_e = P(\Omega)/N = 1/6$

Consider following events:

- odd number: $A_1 = \{1, 3, 5\}$ $\Rightarrow$ $p_1 = 1/2$
- even number: $A_2 = \{2, 4, 6\}$ $\Rightarrow$ $p_2 = 1/2$
- 1 or 6: $A_3 = \{1, 6\}$ $\Rightarrow$ $p_3 = 1/3$
- at least 4: $A_4 = \{4, 5, 6\}$ $\Rightarrow$ $p_4 = 1/2$

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A_1 and A_2 are not independent, they are mutually exclusive:

$$P(A_1 \cap A_2) = P(\emptyset) = 0$$

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A_1 and A_3 are independent:

$$A_1 \cap A_3 = \{1\}$$
$$P(A_1 \cap A_3) = p_e = 1/6 = P(A_1) \cdot P(A_3)$$

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- at least 4: $A_4 = \{4, 5, 6\}$ $\Rightarrow$ $p_4 = 1/2$

A_1 and A_4 are NOT independent:

$$A_1 \cap A_4 = \{5\}$$

$$P(A_1 \cap A_4) = p_e = 1/6 \quad \neq \quad P(A_1) \cdot P(A_4) = 1/4$$

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Probability of A_4 when A_1 is observed:

$$P(A_1 \cap A_4) = 1/6$$

$$P(A_4|A_1) = \frac{P(A_1 \cap A_4)}{P(A_1)} = \frac{1/6}{1/2} = 1/3 \neq P(A_4)$$

Example (2)

Rolling two dice (eg. red and blue): $N = 6 \cdot 6 = 36$ possible outcomes.

Sample space can be best presented as a table:

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

We assume each outcome has the same probability $\Rightarrow p_e = P(\Omega)/N = \frac{1}{36}$

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Sample space can be best presented as a table:

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1			A			B
2			A		B	
3			A	B		
4			$A \cap B$			
5		B	A			
6	B		A			

We assume each outcome has the same probability $\Rightarrow p_e = P(\Omega)/N = \frac{1}{36}$

Events: A - red die shows 3 and B - sum of two dice is 7

$$P(A \cap B) = 1/36 = P(A) \cdot P(B) \quad \Rightarrow \text{are independent}$$

Example (2)

Rolling two dice (eg. red and blue): $N = 6 \cdot 6 = 36$ possible outcomes.

Sample space can be best presented as a table:

	1	2	3	4	5	6
1			A	C		
2			$A \cap C$			
3		C	A			
4	C		A			
5			A			
6			A			

We assume each outcome has the same probability $\Rightarrow p_e = P(\Omega)/N = \frac{1}{36}$

Events: A - red die shows 3 and C - sum of two dice is 5

$$P(A \cap C) = 1/36 \neq P(A) \cdot P(C) = 1/6 \cdot 1/9 \quad \Rightarrow \text{NOT independent}$$

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5			A			
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Events: A - red die shows 3 and C - sum of two dice is 5

$$P(A|C) = P(A \cap C)/P(C) = \frac{1}{36} \ / \ \frac{1}{9} = 1/4$$

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Rolling two dice (eg. red and blue): $N = 6 \cdot 6 = 36$ possible outcomes.

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Events: A - red die shows 3 and C - sum of two dice is 5

$$P(C|A) = P(A \cap C)/P(A) = \frac{1}{36} \; / \; \frac{1}{6} = 1/6$$

Partition of the sample space

It is a set of events A_i $(i = 1 \dots n)$ with the following properties:

- all A_i are mutually exclusive

$$A_i \cap A_j = \emptyset, \qquad \forall i \neq j$$

- they cover the whole sampling space

$$\bigcup_{i=1}^{n} A_i \quad = \quad \Omega$$

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From the two conditions we realize that:

$$\sum_{i=1}^{n} P(A_i) \quad = \quad 1$$

We can be sure that one (and only one) A_i will always take place

Concept of probability

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Total Probability Theorem

Given partition A_i of the sampling space, for any event B we can write

$$P(B) = \sum_{i=1}^{n} P(B \cap A_i) = \sum_{i=1}^{n} P(B|A_i) \cdot P(A_i)$$

Total probability of B can be calculated as a sum over probabilities calculated in separate sub-spaces. Very useful in many cases...

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Example (1)

What is the probability of giving birth to twins in Europe?

We can addressing this problem by combining probabilities for different countries in Europe:

$$P(\text{Twins}) = \sum_{i=1}^{n} P(\text{Twins}|\text{Country}_i) \cdot \frac{N_i}{\sum_{i=1}^{n} N_i}$$

where N_i is the number off all births in country i

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Total probability of B can be calculated as a sum over probabilities calculated in separate sub-spaces. Very useful in many cases...

Example (2)

What is the probability of producing π^+ in e^+e^- annihilation into Z^0:

$$e^+e^- \rightarrow Z^0 \rightarrow \pi^+ + ...$$

We can divide the sampling space, all Z^0 decays, into separate decay channels, $Z^0 \rightarrow f\bar{f}$, where $f = e, \mu, \tau, \nu_e, \nu_\mu, \nu_\tau, u, d, s, c, b$. For some of these channels the answer is known.

eg. $P(\pi^+|Z \rightarrow \nu\bar{\nu}) = 0$

Bayes' Theorem

For events A and B the two conditional probabilities are related:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{and} \quad P(B|A) = \frac{P(B \cap A)}{P(A)}$$

as $B \cap A = A \cap B$ we obtain: Bayes' Theorem

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This can be also written in a more general form:

$$P(A_i|B) = \frac{P(B|A_i)\, P(A_i)}{\sum_{j=1}^{n} P(B|A_j)\, P(A_j)}$$

where A_i is the partition of the sampling space.

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where A_i is the partition of the sampling space.

There is nothing new in this as long as A_i and B belong to the same sampling space...

Bayesian approach

Bayes theorem can be used to generalize the concept of probability.

In particular, one can consider "probability" of given hypothesis H (theoretical model or model parameter, eg. Hubble constant) when taking into known outcome D (data) of the experiment:

$$P(H|D) = \frac{P(D|H)}{P(D)} \cdot P(H)$$

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There are two problems with this approach:

- H can not be considered an event, sampling space can not be defined (no experiment to repeat)
- we need to make a subjective assumption about the "prior" $P(H)$ describing our initial belief in hypothesis H

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For these reasons I rather use term "degree of belief" for the result of the Bayesian procedure applied "outside" the sampling space (not to random events)

Example of Bayesian approach

adapted from G. Cowan, 2011 CERN Summer Student Lectures on Statistics

How much should I worry, if I get a positive test result for virus X?

- Hypothesis: H - I am infected (I am a carrier of X)
- Data: D - test result is positive

How likely is it that I am a carrier of X, what is $P(H|D)$?

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How likely is it that I am a carrier of X, what is $P(H|D)$?

We need to know efficiency and false rate for the test. Let as assume

$$P(D|H) = 0.99 \quad \text{test efficiency}$$
$$P(D|\bar{H}) = 0.01 \quad \text{false positive rate}$$

We can also assume that 1 person per 1000 is an X carrier in our country

$$P(H) = 0.001$$

Example of Bayesian approach

We first use the total probability theorem to calculate the probability of the positive result:

$$\begin{aligned} P(D) &= P(D|H) \cdot P(H) + P(D|\bar{H}) \cdot P(\bar{H}) \\ &= 0.99 \cdot 0.001 + 0.01 \cdot 0.999 = 0.01098 \approx 0.011 \end{aligned}$$

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You can believe that your chances of being infected with X are 9%...

How useful is this information in your opinion?

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This number represents the probability of person with positive test result being infected.

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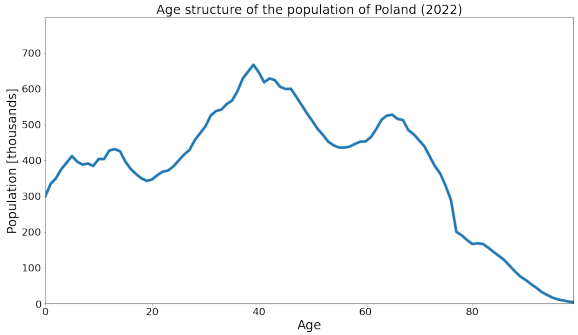
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You can believe that your chances of being infected with X are 9%...

But this can hardly be interpreted as the probability or your infection (not a random variable)...

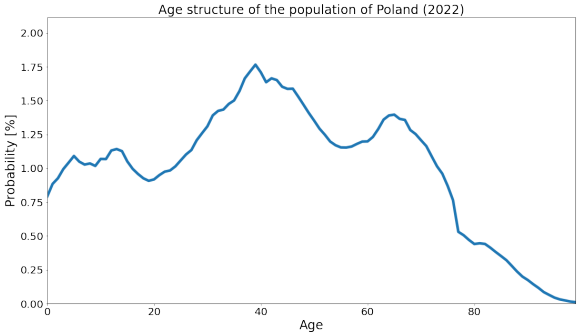
Prior problem

Let us look again at the statistical data on the population of Poland:



Prior problem

Let us look again at the statistical data on the population of Poland:



We can present it as the age probability (for randomly chosen person).

Can we draw any conclusions concerning the life expectation?

Prior problem

To simplify the problem, let us assume that the "survival probability" does not change in time:

$$p(y|b) \quad = \quad s(a)$$

where: b is the year of birth, y is the year of the population census, $a = y - b$ is age.

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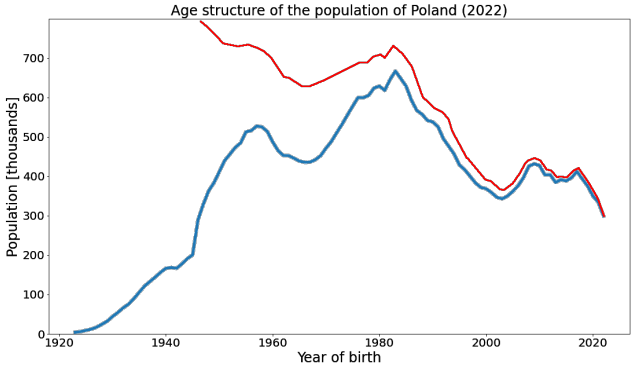
We can apply Bayes' Theorem to relate it to the "survival probability"

$$p(a) \quad = \quad \frac{s(a) \cdot u(y-a)}{\sum_b s(y-b) \cdot u(b)}$$

where $u(b)$ is the birth number distribution. **We need to know it!!!**

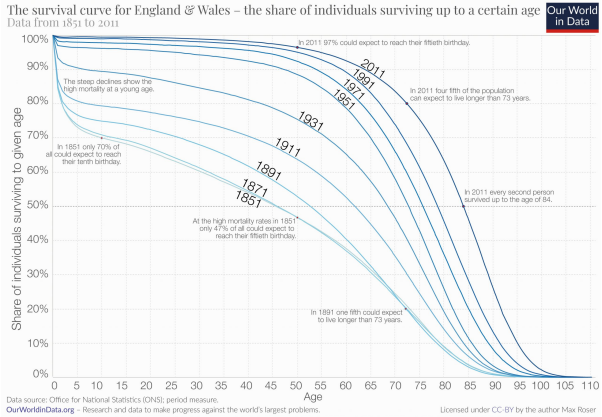
Prior problem

What is needed is the number of births in each year: shown is my guess



Without exact knowledge of this "prior" we can not draw any conclusions...

There are also other reasons, why extraction of survival curve from single population census is not possible. In particular, it depends significantly on the year of birth...



Source: ourworldindata.org

Concept of probability

1 Introduction

2 Basic terms

3 Definition of Probability

4 Properties of Probability

5 Bayes' Theorem

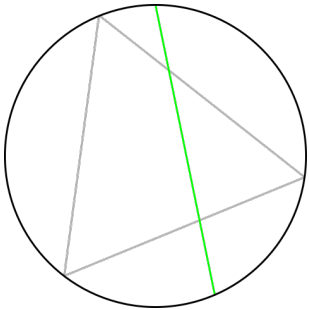
6 Homework

Bertrand paradox

In a circle of radius R an equilateral triangle is drawn. What is the probability that the length of a random chord is greater than the side of the triangle?

Based on the notebook presented at the lecture:

1. present your own solution to the problem: definition of a random chord **different from the three solutions presented at the lecture** (!),
2. prepare the plot showing distribution of the chords generated according to your procedure,
3. calculate the probability considered in the problem.



Solutions should be uploaded until October 28 (Wednesday)